Edwards   Nov 2, '60

air, devel. – small effect at first
aerospace data gathering etc – Cargo.

only guidance & control – Owego
– Kingston

get from Components Group
or field

30 mw about

– ?   100 [illegible]
Tex [illegible] in 100 [illegible]

line length
vs speed of element

1/3 of delay

assume:
1. branching by 3
2. what degree of orderedness?
[illegible] is less than random?   3 or 4
degree of ordering   2 or 3?

cost of wire

1520 ft wire for 4 lots
(2 or 3 [illegible])
20 mils
wire length per [illegible]?

— what about our microwave prog? – what is no. of elements?
what is power?

[illegible] ?
(Twice bandwidth?)
noise level is low --- can they use smaller signals?

1 volt ~100 ohms ≐ 10 mwatt

what percent of lines are terminated " in STRETCH?
" are next to each other
(count no. of terminating resistors)

- are loops com.

→ postulate (on gross basis) { general physical config. / - using this technology or that
{ max distance between devices at diff speeds

1/10 ~~speed~~ time of 1/10 linear dimension 1/1000 bulk

delay times for memories ?
can get

old delay figures } 1/10 μs for [illegible] cores

→ look at those chart

At what point should we express distance on Harvest ?

Priority:

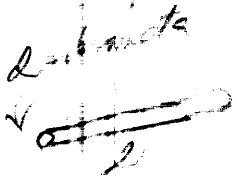
1) Harvest delay
2) Go thru postulated mechanics
3) 1st pass at accommodation

{ 2 or 3 organizations
} hierarchy of memory
(how parallel

→ (read commercial proposal)

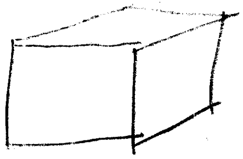
$Vol = L \cdot V_b \cdot f$

If vol. is made up of all wires



$V_w =$ vol of 1 wire $= \frac{1}{4}\pi d^2 \ell$

$V_b = m V_w$ where m = no. wires per block



$Vol = X^3$

$\ell = X$ for random connections

(ie $\frac{1}{3}X + \frac{1}{3}y + \frac{1}{3}z = \frac{3}{3}X$)

$\ell = Xr \leftarrow$ r = "random wire" factor

ℓ how much less than random.

$X^3 = L \cdot m \cdot V_w \cdot f$

$X^3 = L \cdot m \cdot f \cdot \frac{\pi}{4} d^2 X r$

solve for d: $d = \frac{X}{\sqrt{\frac{\pi}{4} L m f r}}$

assume $L = 5 \times 10^4$ elements

$m = 3$

$r = \frac{1}{5}$

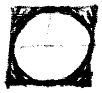
$f = 1.28 = \frac{4}{\pi}$

$X = 1$ ft.

144
12
288
144
1728 cu in

$\therefore \frac{5 \times 10^4}{.1728 \times 10^4} = \underline{\underline{29.}}$ per cu in

assume square packing:



$f = \frac{d^2}{\frac{\pi}{4} d^2} = \frac{4}{\pi} = 1.28$

$$d = \frac{1}{\sqrt{\cancel{\frac{\pi}{4}} \times 5 \times 10^4 \times 3 \times \cancel{\frac{4}{\pi}} \times \frac{1}{5}}} = \frac{12''}{1.72 \times 10^2} = 7.0 \times 10^{-2} = .07''$$

$\frac{1}{16} = .0625$

$\therefore$ wire $\approx \frac{1}{16}''$ inch